

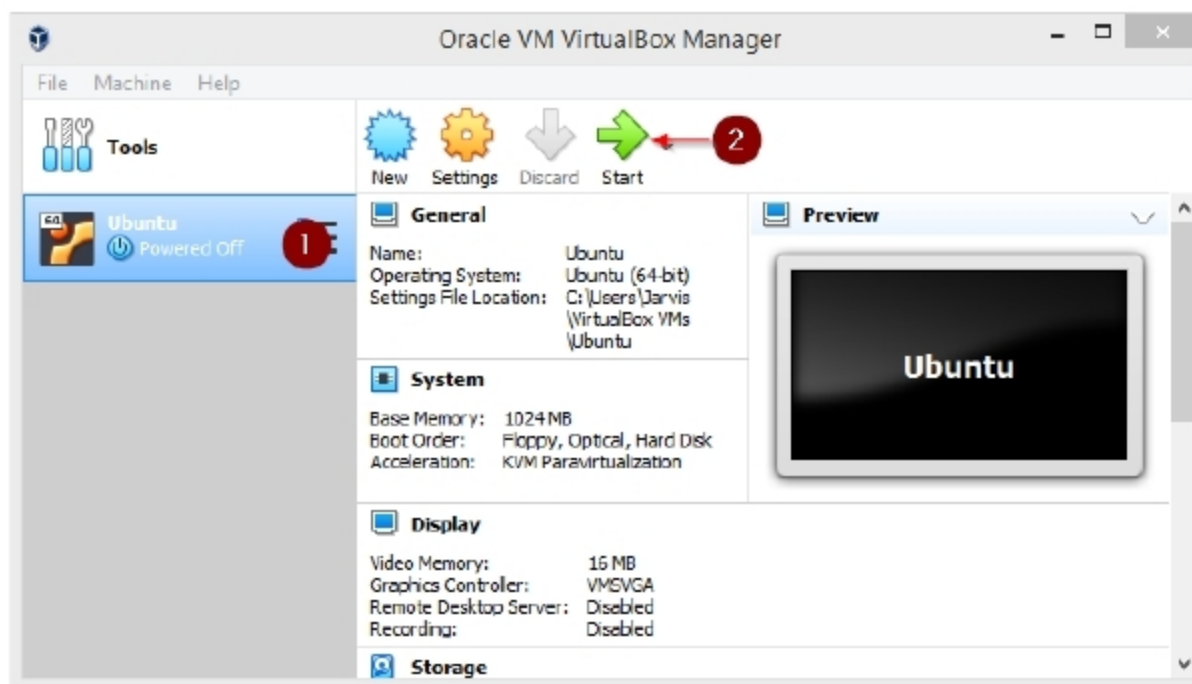


Figure 4: Starting the VM

VM power actions: VirtualBox supports multiple power actions. Let us understand these in detail.

- a) **Power on:** As the term suggests, this starts the VM. To start, right-click on the VM icon and select the *Start* option.
- b) **Power off:** This is a non-graceful turn-off operation. It can cause data loss, so be careful. To power off the VM, right-click on the VM icon and select the *Close->Power off* option.
- c) **Shutdown:** This is a graceful turn-off operation. In this case, the shutdown signal is sent to the guest. To shut down, right-click on the VM icon and select the *Close->ACPI Shutdown* option.
- d) **Reset:** The *Reset* option will turn the VM off and on, respectively. It is different from *Restart*, which is a graceful turn-off operation. To reset the VM, right-click on it and select the *Reset* option.
- e) **Pause:** In this state, the guest releases the CPU but not the memory. As a result, the contents of the memory are preserved when the VM is resumed. To pause the VM, right-click on it and select the *Pause* option.
- f) **Save:** This action saves the current state of the VM and releases the CPU as well as the memory. The saved machine can be started again

in the same state. To save the VM, right click on it and select the *Close->Save State* option.

Removing the VM: Let us explore the steps we need to take to remove a VM. The remove operation can be divided into two parts.

- a) **Unregistering the VM:** This removes the VM from the library, i.e., the VM will get unregistered from VirtualBox and so will not be visible in the VirtualBox manager. To unregister a VM:
 - i) Right-click on VM, and select the *Remove* option.
 - ii) Click on the *Remove Only* option.

Note: To register the VM again, navigate to the *Machine->Add* option of the VirtualBox manager.
 - b) **Deleting the VM:** This action deletes the VM permanently. It will delete the VM configuration files and virtual hard disks. Once performed, this action cannot be undone. To delete the VM permanently, perform the following steps:
 - i) Right-click on VM and select the *Remove* option
 - ii) Click on the *Delete all files* option
- Exporting the appliance:** We can export a VM as an appliance in the Open Virtualization Format (OVF). This comes in a two-file format.
- a) **OVF file format:** In this format, several VM related files are

generated. For instance, there will be separate files for virtual hard disks, configurations and so on. OVF is a platform-independent, efficient, extensible, and open packaging and distribution format for VMs. As it is platform-independent, it allows the import of OVF virtual machines exported from VirtualBox into VMware Workstation Player and vice versa.

- b) **OVA file format:** In this format, all VM related files are archived into a single file and the *.ova* extension is assigned to them. By leveraging this feature, we can create a Golden Image of a VM and deploy multiple instances of it.

To export a VM, perform the steps listed below:

- a) Select a VM from the VirtualBox manager. Navigate to the *File->Export Appliance* option
 - b) Select the VMs to be exported, and click the *Next* button
 - c) Provide the directory location and OVF format version
 - d) Provide the appliance settings and click the *Export* button
- Importing the appliance:** We can create a VM by importing an OVF appliance. To import a VM, perform the following steps:
- a) Open the VirtualBox manager and navigate to the *File->Import Appliance* option
 - b) Select *Virtual appliance* and click on the *Next* button
 - c) Verify the appliance settings and click on the *Import* button.

After this, the new VM will appear in the VirtualBox manager's left pane.

In this article for beginners, we have discussed the benefits of virtualisation, and also looked at how some primitive operations can be performed on virtual machines. **END** 

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